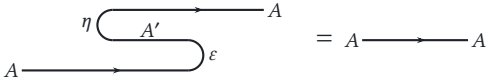
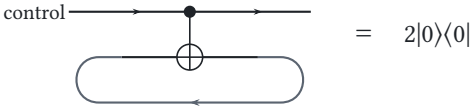
(a) the tensor product: a dual would make the loop of the identity the dimension



the snake equation defines the dual

so $\varepsilon\,\sigma\,\eta = \dim A$, while $|\varepsilon\,\sigma\,\eta| \le \|\varepsilon\|\,\|\eta\| \le 1$
among contractions: no dual once $n \ge 2$

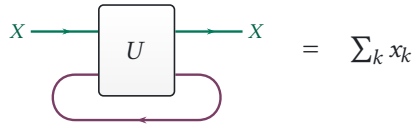
(b) the trace of a factor adds the blocks of one application



tr over the target, a factor

$(I \otimes \langle 0|)\,\mathrm{CNOT}\,(I \otimes |0\rangle) + (I \otimes \langle 1|)\,\mathrm{CNOT}\,(I \otimes |1\rangle)$
$= |0\rangle\langle 0| + |0\rangle\langle 0|$, not a contraction

(c) the trace of a summand adds the exits of the rounds



the loop-back summand Y

$y = Cu + Dy$ implies
$U(u, y) = (\mathrm{Tr}^{Y}(U)u, y)$,
so $\|\,\mathrm{Tr}^{Y}(U)u\| = \|u\|$

(d) the direct sum: the unit and counit through the zero space are zero maps

not the identity of Y: every map into or out of the zero space is zero,
and a unitary into or out of it exists only when $Y = 0$